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Effect of the stability of the poly(gamma-glutamic acid)-ACP dispersion system on biomimetic mineralization of type I collagen†

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Objective: To explore the relationship between the stability of poly(gamma-glutamic acid) (γ-PGA) dispersion systems with γ-PGA of different molecular weights (MWs) and concentrations and type I collagen mineralization. *Methods*: γ-PGA was used as a noncollagenous protein (NCP) analogue to regulate the stability of supersaturated γ-PGA-stabilized amorphous calcium phosphate (PGA-ACP) solutions by changing the γ-PGA MW (2, 10, 100, 200 and 500 kDa) and concentration (400, 500 and 600 μg mL^{-1}). Then, the optical density (OD) at 72 h was measured to determine the PGA-ACP solution stability. Recombinant type I collagen films were mineralized in different PGA-ACP solutions for 3 d and observed *via* transmission electron microscopy (TEM) to confirm the occurrence of intrafibrillar mineralization. The collagen scaffolds were mineralized for 7 d and observed *via* scanning electron microscopy (SEM) to determine the collagen mineralization pattern and degree. X-ray diffraction (XRD), Fourier transform infrared (FTIR) spectroscopy and thermogravimetry (TG) were used to analyse the mineralized collagen scaffold composition. *Results*: The PGA-ACP solutions with γ-PGA of different MWs and concentrations had different stabilities and type I collagen mineralization. Except for the 100 kDa group, which neither stabilized the supersaturated calcium phosphate solution nor induced intrafibrillar mineralization, the groups stabilized the solutions for at least 10 h and induced different intrafibrillar mineralization patterns and degrees. *Conclusion*: In our system, the PGA-ACP solution stability and occurrence of intrafibrillar mineralization are directly correlated. Thus, we suspect that the same correspondence exists in other biomimetic mineralization systems and that a relatively stable supersaturated calcium phosphate solution may be a necessary condition for intrafibrillar mineralization.

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1. Introduction

Caries, which is one of the most common oral diseases with a high prevalence worldwide,[1] involves a chronic and progressive demineralization of dental hard tissues influenced by various factors, mainly bacteria.[2] Classical therapy methods of dental defects like filling and repair often entails the removal of infected dental tissue, and further treatment may be required due to the limited lifespan of artificial materials and the heightened risk of secondary caries,[3–5] thereby imposing both health and financial burdens on patients.[6] With the evolution of minimally invasive treatment approaches for caries in recent years, there has been a growing tendency in clinical practice to preserve more dental hard tissue.[7,8] The utilization of formulations capable of stabilizing caries lesions and promoting remineralization of dental hard tissue has emerged as an effective strategy for both preventing and treating dental caries.[9,10] However, current minimally invasive treatment materials and medications for caries like fluoride and casein phosphopeptide-amorphous calcium phosphate (CPP-ACP), often cannot achieve satisfactory results in dental remineralization[11–13] due to the unique structure of dentin. Dentin, which is a natural organic–inorganic composite material, results from the interaction between type I collagen organic matter and nanohydroxyapatite inorganic minerals and is organized in a hierarchical and orderly manner across seven levels. This intricate arrangement is orchestrated by noncollagenous proteins (NCPs) secreted by odontoblasts, leading to the formation of highly mineralized hard tissue.[14] Based on

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the spatial relationship between collagen fibres and hydroxyapatite, dentin mineralization is categorized into extrafibrillar mineralization occurring outside collagen fibres and intrafibrillar mineralization occurring within collagen fibres. The latter is pivotal in conferring superior mechanical and biological characteristics to natural mineralized tissues.[15,16] Consequently, in the pursuit of dentin remineralization, prioritizing the promotion of intrafibrillar mineralization is paramount. To realize the objective, biomimetic mineralization has emerged as a crucial approach, which means an *in vitro* milieu analogous to the mineralization process within the body is engineered and subsequent mineralization outcomes are attained. However, the intricacies of which remain partially explicated.

Current theories on biomimetic mineralization mechanisms can be broadly classified into two main categories. The first category is rooted in classical crystallization theory, in which calcium and phosphate ions are posited to penetrate the interior of collagen fibres as either ions[17,18] or minute crystals[19] and then undergo crystallization within the collagen matrix and subsequent growth. Conversely, the second category aligns with non-classical crystallization theory, in which the initial formation of a metastable mineralized precursor, known as amorphous calcium phosphate (ACP),[20,21] outside the collagen fibre is proposed, after which the precursor undergoes crystallization within the collagen matrix. The experimental identification of ACP derived from polyelectrolyte-stabilized mineralization solutions[22] provides additional empirical support for this theory. And the process of forming ACP and inhibiting the precipitation of minerals is called stabilization. Both classical and non-classical crystallization theories currently emphasize the pivotal role of the driving forces responsible for infiltration of mineral ions/mineralized precursors into collagen fibres, and numerous theories have emerged to elucidate this phenomenon. For example, the capillary action between liquid phases in Gower's polymer-induced liquid process (PLIP) theory,[23] Coulomb gravity between positively and negatively charged regions,[22,24–27] and osmotic pressure–charge double-balance force in recent study of Niu *et al.*[28] However, regardless of the theoretical framework employed, each theory can elucidate only a portion of the biomimetic mineralization phenomenon, with inherent limitations and complementary aspects.[29] Before considering the specific driving force mechanisms, we conducted a comprehensive review of these theories and observed that non-classical crystallization theories uniformly rely on the presence of metastable mineralization precursors. By utilizing polyelectrolytes to mimic the functions of NCPs, these theories demonstrate their efficacy in stabilizing mineral ions to form mineralization precursors. Conversely, classical crystallization theory often presupposes the existence of free mineral ions in solution or mineral crystals of a size conducive to penetrating collagen fibres. Building upon this premise, we propose the following hypothesis: provided that mineral ions can maintain a stable or metastable state for a period, they could infiltrate the interior of collagen fibres under the templating influence of the fibres themselves, thereby undergoing intrafibrillar mineralization.

To evaluate this conjecture, we endeavoured to modulate the stability of supersaturated calcium phosphate solutions and investigate the correlation between their stability and the occurrence of mineralization within collagen fibres. Based on a literature review, we observed that polyglutamic acid (PGA) shares similarities with polyaspartic acid (PAsp) as an anionic polymer. Theoretically, it can function as an NCP analogue capable of chelating calcium ions to stabilize supersaturated calcium phosphate solutions and facilitate subsequent mineralization within collagen fibres. However, empirical evidence reveals that L-PGA exhibits limited efficacy in stabilizing mineralization solutions and fails to form stable ACP structures, consequently impeding its ability to guide mineralization within collagen fibres.[30,31] Nevertheless, certain investigations have indicated that as the ratio of enantiomers in PGA approaches parity, the stability of supersaturated calcium phosphate solutions increases.[32] Furthermore, the application of PGA enantiomers formed from glutamic acid units *via* γ-amide bond connections can effectively guide remineralization both between demineralized dentin tubules and within dentin tubules.[33] This observation underscores the correlation between the stability of PGA-stabilized mineralization fluid and mineralization within collagen fibres. Based on the findings of previous researchers, manipulation of the molecular weight and concentration of polyelectrolytes can modulate the stability of mineralization solutions.[22,34,35] Therefore, in this study, cost-effective γ-PGA with a broad molecular mass range was used as a surrogate for NCPs to assess the stability of diverse PGA-ACP dispersion systems and their impact on the mineralization of collagen fibres. Subsequently, the correlation between the stability of supersaturated calcium phosphate solutions and the occurrence of intrafibrillar mineralization was analysed.

Considering the semipermeable characteristics of collagen fibres,[36,37] PGA materials of various molecular weights were investigated in this study include 2, 10, 100, 200 and 500 kDa. Then, the optical density (OD) values at 72 h were measured to determine the stability of the PGA-ACP solutions. Furthermore, graphical representations of the pH curves for each solution over time were generated, and the morphological attributes of ACP were meticulously examined utilizing transmission electron microscopy (TEM). Recombinant type I collagen films were mineralized in different PGA-ACP solutions for 3 d and observed *via* TEM to confirm the occurrence of intrafibrillar mineralization. The collagen scaffolds were mineralized for 7 d and observed *via* scanning electron microscopy (SEM) to determine the pattern and degree of collagen mineralization. Meanwhile, X-ray diffraction (XRD), Fourier transform infrared (FTIR) spectroscopy and thermogravimetry (TG) were used to analyse the composition of the mineralized collagen scaffold. Ultimately, a connection between the stability of the PGA-ACP dispersion system and the occurrence of intrafibrillar mineralization is explicitly observed, which might provide a new conceptual avenue for existing theories and offer fresh insights for future investigations into biomimetic mineralization.

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2. Materials and methods

2.1 Construction of biomimetic mineralization solutions

$CaCl_2 \cdot 2H_2O$ (Macklin, Shanghai, China) and K_2HPO_4 (Macklin, Shanghai, China) were dissolved in HEPES buffer to form supersaturated solutions with concentrations of 9 mM Ca^{2+} and 4.2 mM HPO_4^-, and the pH was adjusted to approximately 7.4 with NaOH. PGA of five different MWs (2 kDa, 10 kDa, 100 kDa, 200 kDa and 500 kDa (Nanjing Saitaisi Biotech, China)) was used to stabilize ACP. Each PGA was dispersed in the calcium solution at different concentrations (800, 1000 and 1200 μg mL^{-1}) and then mixed with an equal volume of the phosphate solution. The concentrations of Ca^{2+} and HPO_4^- in the final PGA-ACP dispersion were 4.5 mM and 2.1 mM, and the concentrations of PGA were 400, 500 and 600 μg mL^{-1}, respectively. Each group was named "MW-final concentration in the mineralization dispersion". For instance, a mineralization dispersion with 200 kDa PGA at a concentration of 400 μg mL^{-1} was called "200k–400". Due to the elevated levels of calcium and phosphorus ions present in the solution, precipitation can theoretically occur rapidly in the absence of γ-PGA. Consequently, when PGA or other materials is introduced to prolong the stabilization period and maintain the solution in a metastable condition, it is referred to as a supersaturated calcium phosphate (CaP) solution. Similarly, 1.8 kDa poly(acrylic acid) (PAA), a polymer with enough carboxyl to stabilize supersaturated CaP solutions, was dissolved in the CaP solution at a final concentration of 50 μg mL^{-1} as a positive control, and the mineralization solution without any stabilizing polymers was used as a blank control.

2.2 Stability assessment of the PGA-ACP dispersions

The OD value of the mineralization solutions was determined *via* absorbance experiments to assess the turbidity of the mineralization solutions within each experimental group and concurrently evaluate the stabilizing effect of PGA with various molecular weights and concentrations on ACP. A 300 μL volume from each mineralization solution group was dispensed into a 96-well plate. Simultaneously, deionized water was dispensed into the same plate. Subsequently, an OD measurement was conducted at a wavelength of 650 nm using a multifunctional microplate reader. Measurements were conducted at regular intervals, with the test interval set to every 2 h within the initial 0 to 24 h period and every 12 h within the subsequent 24 to 72 h timeframe. Each well was measured three times, and the mean value was calculated. The OD of the deionized water served as the baseline. The final OD was determined by subtracting the baseline OD of the deionized water from the OD of each mineralization solution group. All the OD values are used to get a detailed precipitation curve and to calculate the precipitation half-time, $t_{1/2}$, of each group. Curves of OD as a function of time (t) were fitted to a logistic function:

$$\mathrm{OD}(t) = A2 + (A1 - A2)/(1 + (t/t_{1/2})^p)$$

where $A2$ is the initial OD value at $t \to 0$, $A1$ is the final OD value at $t \to \infty$, $t_{1/2}$ is the midpoint of the OD curve, and p is a constant of each group. The lager the $t_{1/2}$, the longer the stable time of PGA-ACP. The curve were iterated using the Levenberg–Marquardt optimization algorithm until the fit converged to a Chi-sqr tolerance of 1×10^{-9}.

A pH indicator was used to assess the pH of each set of mineralization solutions, with measurements taken at two-hour intervals during the initial 0 to 24 h period and every 12 h thereafter from 24 to 72 h. Each measurement was replicated three times, and the mean value was calculated to ensure accuracy. Then the pH value are smoothed to get a continuous curve. According to the turning points of pH curve, the time points of TEM observation are determined and status of the mineralization precursor were determined by selected area electron diffraction (SAED).

The particle size distribution of each set of mineralization solutions was determined by employing a particle size analyser coupled with dynamic light scattering (DLS). Additionally, the zeta potential of the nanoparticles suspended within each set of mineralization solutions was quantified utilizing a zeta potential analyser. Measurements were conducted at 24, 48, and 72 h following the initiation of mineralization, with each measurement replicated three times to ensure precision. The mean values were calculated from the collected data, and the results are presented as the mean ± standard deviation.

2.3 "On-grid" self-assembly and mineralization of type I collagen

Type I collagen powder derived from bovine skin (Sigma-Aldrich, St Louis, MO, USA) was dissolved in 0.1 M acetic acid (pH = 3) at a concentration of 0.2 mg mL^{-1}, and then, the type I collagen solution was placed in a refrigerator at 4 °C overnight for complete dissolution. Formvar-and-carbon-coated 400-mesh Ni TEM grids were floated upside down in a small Petri dish containing the collagen solution. The collagen solution was neutralized with 1%–1.5% ammonia solution for 4 h until the pH increased to 8, after which the mixture was incubated at 37 °C for 3 d after sealing. The TEM grids were treated with 0.3 M 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide (EDC, Macklin, Shanghai, China)/0.06 M *N*-hydroxysuccinimide (NHS, Macklin, Shanghai, China) solution at ambient temperature for 4 h to further stabilize the structure of the collagen fibres. Then, the TEM grids containing thin layers of recombinant type I collagen were randomly allocated to each mineralization dispersion group and mineralized at 37 °C for 72 h.

2.4 Preparation and mineralization of three-dimensional type I collagen scaffolds

The tails of fresh 8 week-old SD rats were immersed in 75% ethanol solution for 15 min and washed with deionized water. Then, the white tail tendons were extracted and rinsed with phosphate-buffered saline (PBS). The samples were immersed in a 0.05 mol L^{-1} Tris-HCl-NaCl (pH = 7.4) buffer solution for 24 h at 4 °C. Then, the rat tail tendons were dissolved in 0.3 M acetic acid and stirred with a magnetic stirrer for 3 d. The collagen solution was centrifuged for 30 min at 3000 rpm at 4 °C

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in a refrigerated centrifuge, and the supernatant was collected and placed in a dialysis bag. The dialysis bag was immersed in 0.02 M K_2HPO_4 solution for 5–7 d to obtain the collagen gel. The collagen gel was freeze-dried in a vacuum freeze-dryer and pressed with a cylindrical mould to prepare a collagen scaffold with a diameter of 8 mm and a thickness of 2 mm. Then, the collagen scaffolds were randomly dispersed in a mineralization dispersion in each group and mineralized at 37 °C for 7 d. The mineralization solution was changed every 2 d. After 7 d, the scaffolds were removed, washed with deionized water, placed at −80 °C overnight, and then freeze-dried by a vacuum freeze dryer.

2.5 Transmission electron microscopy (TEM)

After 72 h of mineralization, TEM grids containing reconstituted collagen were randomly selected for observation. The TEM grids were rinsed with DI water and air dried for observation using a transmission electron microscope (TEM1400; JEOL, Japan and JEM-2100F; JEOL, Japan) operated at 120 kV in bright field mode. The mineral properties of the collagen fibres were determined by SAED.

2.6 Scanning electron microscopy (SEM)

A three-dimensional collagen scaffold sample was taken from each group after 7 d of mineralization, and the microstructure of the collagen scaffold was imaged under a cold field emission scanning electron microscope (Regulus 8230, HITACHI, Japan) operated at an accelerating voltage of 15 kV. The energy dispersive spectrometer (EDS) analysis of suitable group was did to confirm the intrafibrillar mineralization.

2.7 X-ray diffraction (XRD)

The crystal structure of the collagen scaffolds was analysed with an X-ray powder diffractometer (Empyrean, PANalytical B. V., Almelo, Netherlands). The samples were scanned with CuKα X-ray radiation at 40 kV and 35 mA. The incident angle was 15°, and the detector position was fixed at 30°, which covered a 2θ = 15°–45° range.

2.8 Fourier transform infrared (FTIR) spectroscopy

The collagen scaffolds of each group were tested by FTIR spectrometry (Nicolet 6700-Contiumm, Thermo Scientific, Waltham, MA, USA). Each spectrum was obtained by averaging the signals of 32 scans with a resolution of 2 cm^{-1}, and the wavenumber ranged from 400 to 4000 cm^{-1}.

2.9 Thermogravimetric and derivative thermogravimetric (TG/DTG)

The mineral content of the collagen scaffolds was determined by TG/DTG (TG-209, Netzsch, Germany). The temperature was increased from 30 to 800 °C at a heating rate of 5 °C min^{-1} under nitrogen. The acquired data were utilized to generate a mass-temperature curve using Origin software.

3. Results

3.1 Characterization of the PGA-ACP dispersions

This investigation illustrates the stability of supersaturated CaP solutions augmented with PGA of various concentrations and diverse molecular weights. This was accomplished through assessment of the OD (Fig. 1A–E) and the precipitation half-time, $t_{1/2}$, of each group (Table 1). What's more, zeta potential (Fig. 1P–T) of the PGA-ACP solutions and the nanoparticle size of the mineralized precursor stabilized by PGA (Fig. 1K–O) can be additional support. Theoretically, pH value is a indirect method for estimating the state of ACP particle and subsequently verified by the observation of TEM microscope. In general, PGA with different MW are capable of conferring distinct stabilization abilities on supersaturated CaP solutions at equivalent concentrations. Within the context of identical MW, seemingly there is a trend that an increased concentration correlates with a more marked enhancement ability to stabilization, albeit the magnitude of this discrepancy is significant in 200 kDa groups only. Specifically, the MW of PGA arranged in ascending order of stabilization ability are as follows: 100 kDa < 2 kDa < 10 kDa ≤ 500 kDa < 200 kDa.

Using the PAA-ACP solution at 1.8 kDa and 50 μg mL^{-1} as the positive control, before jumped sharply the OD values of the PGA-ACP solutions in the 2 kDa, 10 kDa, 200 kDa, and 500 kDa groups were closed to the control group and remain relatively stable. In contrast, in the 100 kDa group, the OD value began to increase after the second hour and surpassed that of the positive control group. Over the 0–72 h timeframe, the OD values of the experimental groups were consistently lower than those of the negative control group lacking electrolyte stabilization. According to the calculations of $t_{1/2}$ (Table 1), except for 100 kDa PGA, which failed to stabilize the supersaturated CaP solutions, other PGA in this study exhibited the capability to stabilize the supersaturated CaP solutions at all concentrations for a minimum duration of nearly 10 h.

To determine the effect of PGA on the supersaturated CaP solution, a particle size analysis was conducted utilizing a particle size analyser coupled with DLS technology to assess the nanoparticle size in each group at three time points: 24, 48, and 72 h (Fig. 1K–O). Compared to those in the negative control group and the 100 kDa group, the nanoparticle diameters in the other experimental groups were notably smaller and remained relatively stable throughout the experiment. The positive control group containing PAA exhibited the smallest and most stable nanoparticle diameter within the 72 h timeframe. The nanoparticle sizes within the 100 kDa group (Fig. 1M) ranged from 1000 to 4000 nm, while in the 10 kDa (Fig. 1L) and 200 kDa (Fig. 1N) groups, the nanoparticle sizes ranged from 150 to 400 nm. Moreover, in the 2 kDa (Fig. 1K) and 500 kDa (Fig. 1O) groups, the nanoparticle sizes ranged from 500 to 800 nm. The zeta potential values (Fig. 1M–P) for all the solutions in the experimental groups were found to be negative, which is a characteristic attributed to the negative charge of PGA within the solution. Furthermore, the absolute values of the zeta potential in both the experimental and posi-

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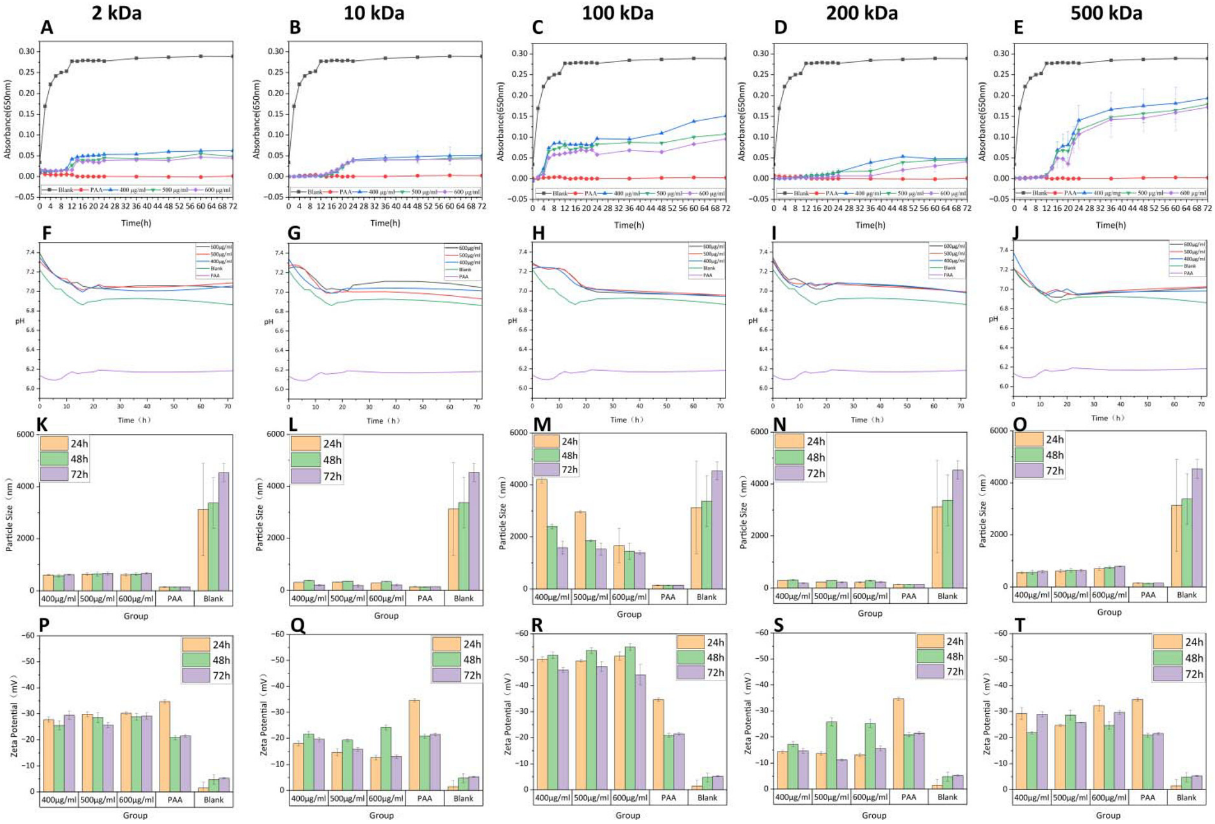

Fig. 1 OD (A–E), pH (F–J), nanoparticle size (K–O) and zeta potential (P–T) of the PGA-ACP dispersion systems. The relative molecular masses of PGA are 2 kDa (A, F, K and P), 10 kDa (B, G, L and Q), 100 kDa (C, H, M and R), 200 kDa (D, I, N and S) and 500 kDa (E, J, O and T).

Table 1 Precipitation half-time ($t_{1/2}$, h) for PGA-ACP solutions with PGA of different MWs and concentrations

	PGA concentration		
PGA MW (kDa)	400 μg mL^{-1}	500 μg mL^{-1}	600 μg mL^{-1}
2	11.5	12.3	12.4
10	20.2	20.1	18.8
100	5.8	5.4	5.7
200	27.1	30.8	48.0
500	19.6	22.2	22.4

tive control groups were notably greater than those in the negative control group.

Simultaneously, to indirectly estimate the status of ACP particles and analyze the reaction process of the PGA-ACP dispersion system, the pH of the PGA-ACP solution was assessed (Fig. 1F–J), revealing that the pH of the positive control group remained relatively constant at approximately 6.1. In contrast, the pH of the negative control group decreased from 7.3 to approximately 6.9 within the initial 0–16 h and subsequently remained stable from 16–72 h. The pH of the 100 kDa group stabilized at approximately 7.3 within the initial 0–12 h and subsequently decreased to approximately 7.0 within the 12–20 h timeframe before stabilizing. Generally, the 2 kDa, 200 kDa, 500 kDa, 10k–400 and the negative control group have similar shape with one turning point on the pH curve, while the 100 kDa, 10k–600 and 10k–500 groups have similar shape with two turning points.

Considering the shape of pH curve and the different ability of stabilization, 2k–600, 10k–600 and 100k–600 groups were thought as the representatives to directly observe the ACP particle under TEM microscope (Fig. 2E–N and ESI Fig. 1†). Different from the negative control group (ESI Fig. 1F†) whose ACP stage is too short to observe, in the 100k–600 group, ACP particle can be seen in the 6th hour (Fig. 2L), whose Ca/P is 1.44–1.86 (Fig. 2D), higher than other groups. At other period, the image appears as a filamentous substance intervening condition between ACP and HA (Fig. 2M–P). In the 10k–600 group, ACP particle can be seen before 14 hours (Fig. 2F–I) and the Ca/P is 1.24–1.36 (Fig. 2B). Whereafter, the characteristic diffraction rings (002) and (211) can be observed at 20th hour (Fig. 2J). In the 2k–600 group (ESI Fig. 1F–H†), ACP particle can be seen before the turning point, and similar filamentous substance from the turning point to later.

3.2 Effect of the MW and concentration of PGA on collagen mineralization

In the present study, the fabricated two-dimensional collagen model was subjected to mineralization within PGA-ACP solutions of diverse molecular weights and concentrations for 72 h. Subsequently, the mineralization process occurring

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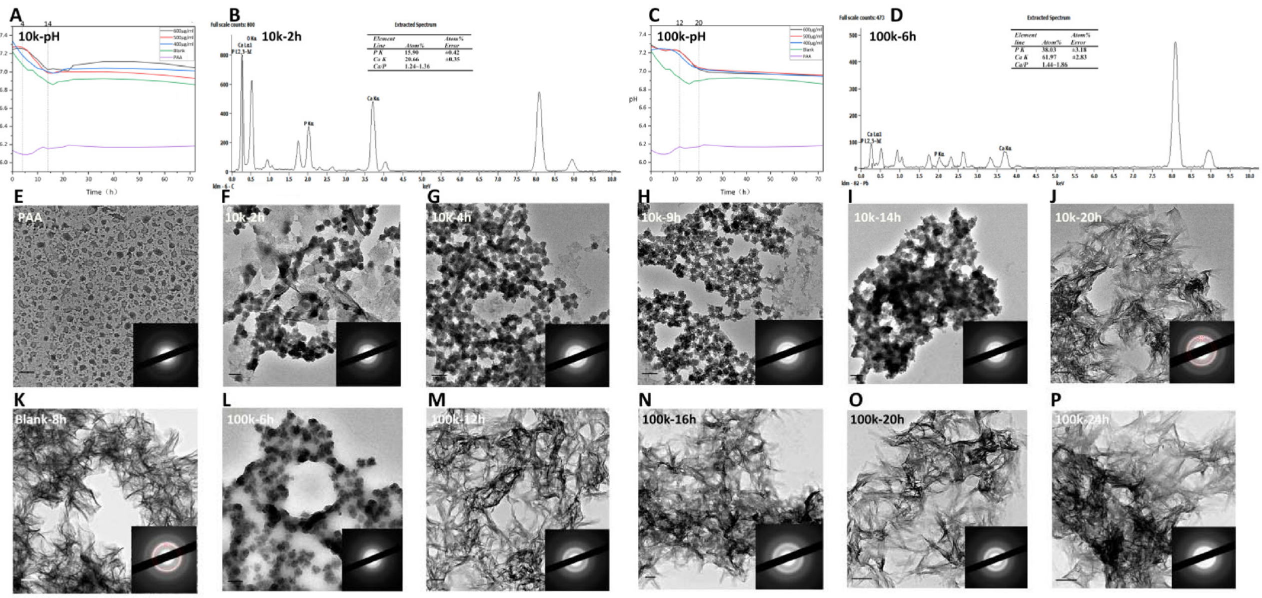

Fig. 2 pH curves of 10 kDa group (A) and 100 kDa group (C) with two turning points. TEM imagines and SAED imagines of CaP solutions in 10k–600 group (F–J) and 100k–600 group (L–P), whose time points of observation depends on the turning points of pH curve. (B) The EDS graph of graph (F); (D) the EDS graph of graph (L); (E) the TEM imagine of PAA group at 4 hours. (K) the TEM imagine of Blank group at 8 hours. Bar: 100 nm.

within the collagen fibres was observed *via* TEM (Fig. 3). Except for the 100 kDa group, the mineralized collagen fibres across all the experimental groups exhibited traits analogous to those of collagen fibres with intrafibrillar mineralization. These traits typically manifest as an augmented electron density and an enlarged fibre diameter, albeit aggravated by the distinct molecular weights and concentrations of PGA. Notably, the mineralization attributes varied depending on these parameters. In the 2 kDa PGA group (Fig. 3A–C), conspicuous mineral deposition within the collagen fibres was observed, alongside mineral deposition on the external surface of the collagen fibres, giving rise to extrafibrillar mineralization. This phenomenon was also observed in both the 10 kDa PGA group (Fig. 3D–F) and the 500 kDa PGA group (Fig. 3M–O) and was particularly evident in the 500k–400 group (Fig. 3M). In the 500k–400 group, where minerals (indicated by red arrows) were discernible, substantial accumulation of collagen fibres was observed on the surface, which were interconnected to form a brush-like configuration. In the 200 kDa PGA group (Fig. 3J–L), conspicuous intrafibrillar mineralization was evident, particularly in the collagen mineralization of the 200k–400 group (Fig. 3J) and the 200k–500 group (Fig. 3K), wherein the characteristic D-period of collagen was discernible in TEM observations, manifesting as an alternating light and dark striated structure. This phenomenon arises from the orderly deposition of minerals within the pore region of the collagen fibres. Consequently, this mineralized collagen is referred to as hierarchical intrafibrillar mineralized collagen. Meanwhile, there was an absence of discernible mineralization along the periphery of collagen fibers within the two groups, indicative of a incomplete mineralization. In each sample of the 100 kDa group (Fig. 3G–I), mineralized crystals were observed to be evenly deposited across the entire field of view, with collagen fibres exhibited as a background with a low electron density, devoid of any discernible mineral deposition within their structure. In the positive control group incorporating PAA (ESI Fig. 2A†), notable augmentation of the electron density within the collagen fibres was evident, albeit without the striated structure observed in the 200 kDa group, indicating the ability of the positive control group to promote intrafibrillar mineralization without attaining hierarchical intrafibrillar mineralization. Conversely, in the negative control group (ESI Fig. 2B†), extensive disorderly mineral deposition dominated the field of view, and the presence of collagen fibres was difficult to discern.

In this study, three-dimensional collagen scaffolds derived from rat tail collagen were subjected to mineralization within PGA-ACP solutions featuring diverse molecular weights and concentrations over a duration of 7 d. After dehydration in ethanol/milliQ water[38] and freeze-drying, the nature and magnitude of mineralization were meticulously examined utilizing SEM (Fig. 4). SEM revealed concurrent extrafibrillar and intrafibrillar mineralization of collagen fibres in the 2 kDa, 10 kDa, 200 kDa and 500 kDa groups, while in the 100 kDa group, no intrafibrillar mineralization could be observed. Within the 100 kDa PGA experimental group (Fig. 4G–I), only extensive disordered mineral deposition was observable. Collagen fibres appeared to be obscured by deposits, rendering them challenging to discern within the field of view, exhibiting analogous behaviour to that of the negative control group (ESI Fig. 3B†). The prominent intrafibrillar mineralization observed in the 200 kDa groups (Fig. 4J–L), resulted in an collagen fibre diameter and a smooth surface of collagen fibre bundle around which scaly (200k–400) or spherical (200k–500 and 200k–600)

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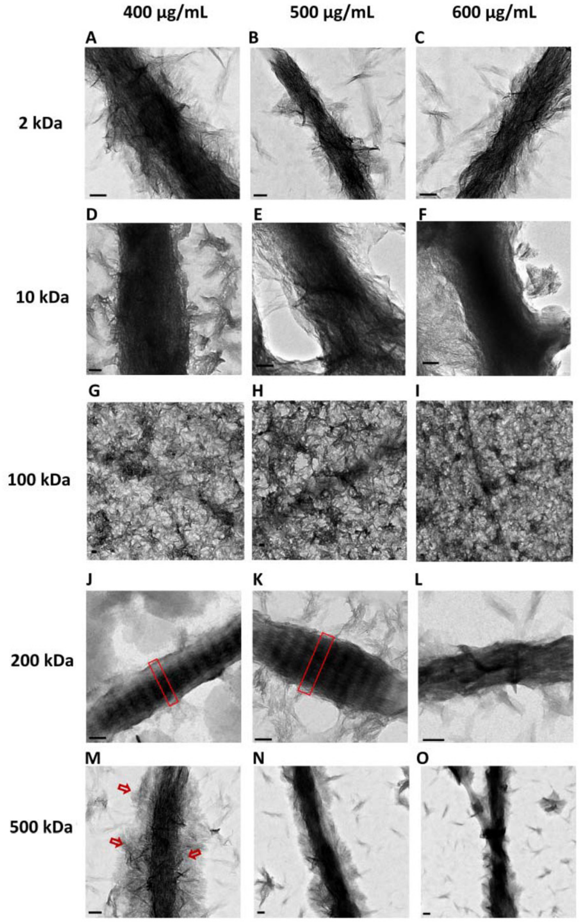

Fig. 3 TEM images of the two-dimensional collagen model in each experimental group after 3 d of mineralization. (A–C): 2 kDa–400 μg mL^{-1}, 2 kDa–500 μg mL^{-1}, 2 kDa–600 μg mL^{-1}; (D–F) 10 kDa–400 μg mL^{-1}, 10 kDa–500 μg mL^{-1}, 10 kDa–600 μg mL^{-1}; (G–I) 100 kDa–400 μg mL^{-1}, 100 kDa–500 μg mL^{-1}, 100 kDa–600 μg mL^{-1}; (J–L): 200 kDa–400 μg mL^{-1}, 200 kDa–500 μg mL^{-1}, 200 kDa–600 μg mL^{-1}; (M–O): 500 kDa–400 μg mL^{-1}, 500 kDa–500 μg mL^{-1}, 500 kDa–600 μg mL^{-1}. Red box in (J–K) horizontal stripe structure; red arrow in (M) extrafibrillar mineralization; scale bar: 100 nm.

mineralized particles outside collagen fibres can be seen sporadically. In the residual experimental groups, simultaneous intrafibrillar mineralization and extrafibrillar mineralization were also observed. Obvious scaly mineral deposits could be observed on the surface of collagen fibre bundle, and the loose collagen fibre structure observed in pure collagen (ESI Fig. 3A†) is disappeared. When we magnify the collagenous bundle structure in experimental groups, mineralized collagen with full and smooth surface as marked in Fig. 5 with red circle can be clearly different from the no mineralized collagen with thin form and transverse stripe, which were marked with black circle. In particular, there were also full fibres with transerve stripes on the surface in some groups (ESI Fig. 4–6†), which may be related to the hierarchical intrafibrillar mineralized and incomplete mineralization of collagen fibres.

To prevent the interference of extrafibrillar mineralization deposited on the surface of collagen, 200k–500 group was selected to do a EDS analysis (Fig. 6). Obviously, most collagen is lack of minerals compared with extrafibrillar mineralized particles. But when the point analysis focused on collagen regions lacking minerals, calcium and phosphorous were also observed with ratio of Ca/P is 1.54.

3.3 Characterization of minerals

TEM revealed hierarchical intrafibrillar mineralization in both the 200k–400 group (Fig. 7A) and the 200k–500 group (Fig. 7C), which were subjected to subsequent SAED analysis. SAED analysis revealed distinctive diffraction rings corresponding to the (002) and (211) planes of hydroxyapatite (HA) crystals in both groups (Fig. 7B and D). Moreover, SAED analysis was conducted on the experimental groups with intrafibrillar mineralization (ESI Fig. 7†), revealing the presence of characteristic (002) and (211) diffraction rings of HAp crystals across these groups.

XRD analysis was employed to investigate the mineral composition of mineralized collagen samples (Fig. 7E–J). The findings revealed the presence of characteristic diffraction peaks corresponding to the (002) and (211) planes of the HAp crystals in the collagen scaffolds of all the experimental groups, indicating HAp crystal formation. However, the intensities of the (002) and (211) diffraction peaks were weaker in the 10 kDa group (Fig. 7F) and the 100 kDa group (Fig. 7G), especially in the latter. In contrast, the diffraction peaks of HAp crystals in the 200 kDa group (Fig. 7H) exhibited relatively strong intensities. Notably, no discernible characteristic diffraction peaks were observed in the pure collagen group (Fig. 7J).

FTIR was utilized to elucidate the chemical composition of the collagen scaffolds in each group (Fig. 7K–P). Characteristic absorption peaks corresponding to amide A and B groups, as well as amide I, II, and III bands, were observed in all the groups. Notably, distinctive absorption peaks of the phosphate group were evident in each experimental group, represented by the υ_3PO_4 stretching vibration peak at 1020 cm^{-1} and the υ_1PO_4 stretching vibration peak at 960 cm^{-1}. HAp exhibited υ_3CO_4 stretching vibration peaks at 1410 cm^{-1} and 870 cm^{-1}, akin to those of natural bone minerals, alongside υ_2CO_4 bending vibration peaks produced by carbonate. In the pure collagen group (Fig. 7P), the residual HPO_4^- introduced during collagen scaffold preparation was not completely eliminated during dialysis, as indicated by the υ_3PO_4 stretching vibration peak.

3.4 Quantification of mineralization

Thermogravimetric analysis (TGA) was conducted on the freeze-dried three-dimensional collagen scaffold to determine the mineral content (Fig. 8A–E), and DTG curves were then generated to compare the compositions of mineralized collagen across each experimental group (Fig. 8F–J).

TGA revealed that the mineral contents of the PAA positive control group, polymer-free blank control group, and pure collagen group were 40.22%, 29.10%, and 21.44%, respectively. Post-mineralization, the mineral contents across the experi-

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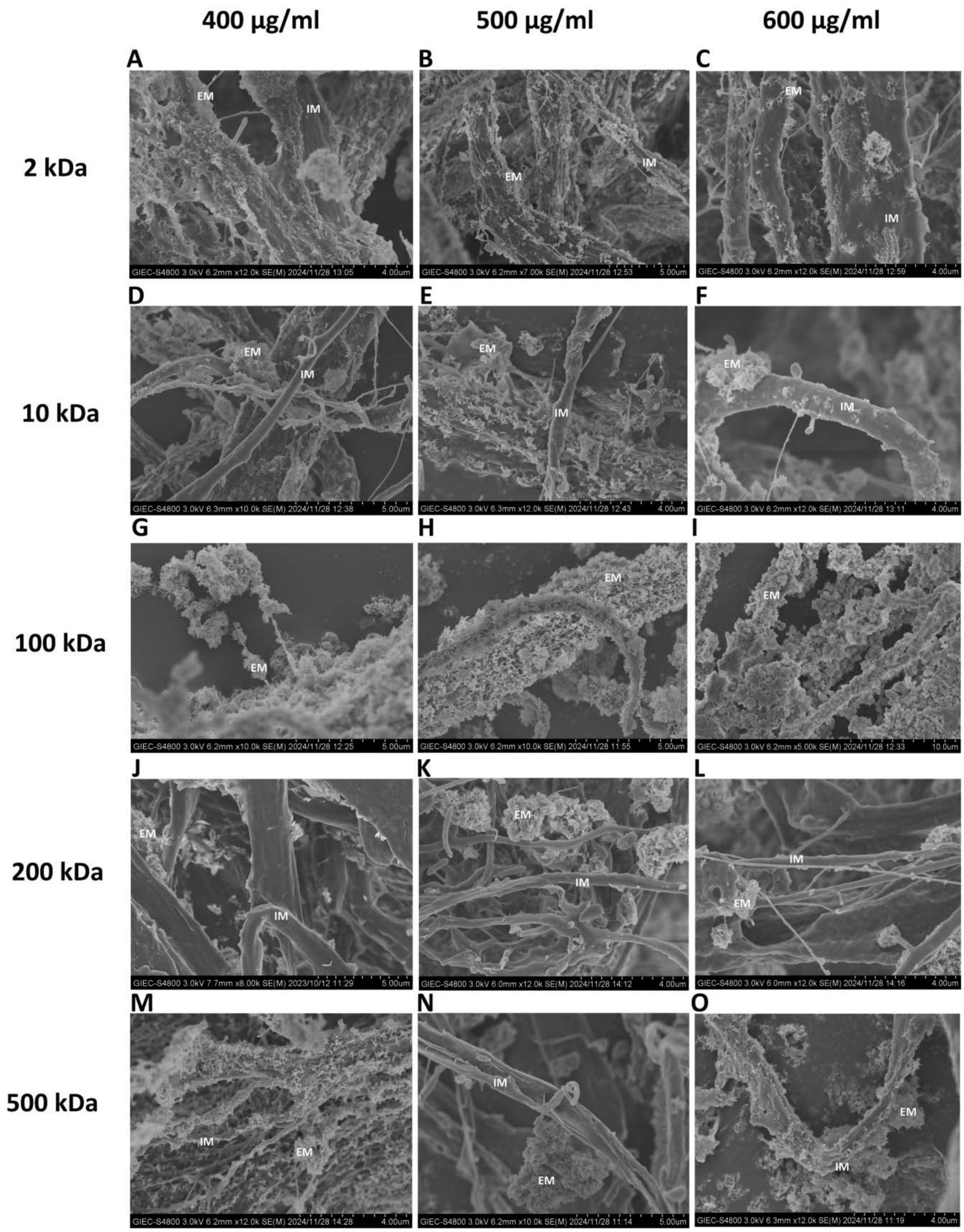

Fig. 4 SEM images of the mineralized three-dimensional collagen scaffolds in each experimental group after 7 d. (A–C): 2 kDa–400 μg mL^{-1}, 2 kDa–500 μg mL^{-1}, 2 kDa–600 μg mL^{-1}; (D–F) 10 kDa–400 μg mL^{-1}, 10 kDa–500 μg mL^{-1}, 10 kDa–600 μg mL^{-1}; (G–I) 100 kDa–400 μg mL^{-1}, 100 kDa–500 μg mL^{-1}, 100 kDa–600 μg mL^{-1}; (J–L) 200 kDa–400 μg mL^{-1}, 200 kDa–500 μg mL^{-1}, 200 kDa–600 μg mL^{-1}; (M–O) 500 kDa–400 μg mL^{-1}, 500 kDa–500 μg mL^{-1}, 500 kDa–600 μg mL^{-1}. IM: intrafibrillar mineralization; EM: extrafibrillar mineralization; scale bar: 1 μm.

mental groups ranged from 18.10% to 45.53% (Fig. 8A–E). Notably, the 200k–500 group exhibited the highest mineral content (Fig. 8D), surpassing that of the PAA positive control group. Conversely, the 100k–500 group displayed the lowest mineral content (Fig. 8C), which was marginally lower than that of the pure collagen group. The mineral contents in the 2 kDa and 500 kDa groups (Fig. 8A and E) exceeded that in the blank control group, while the 10 kDa and 100 kDa groups (Fig. 8B and C) demonstrated lower mineral contents compared to the blank control group.

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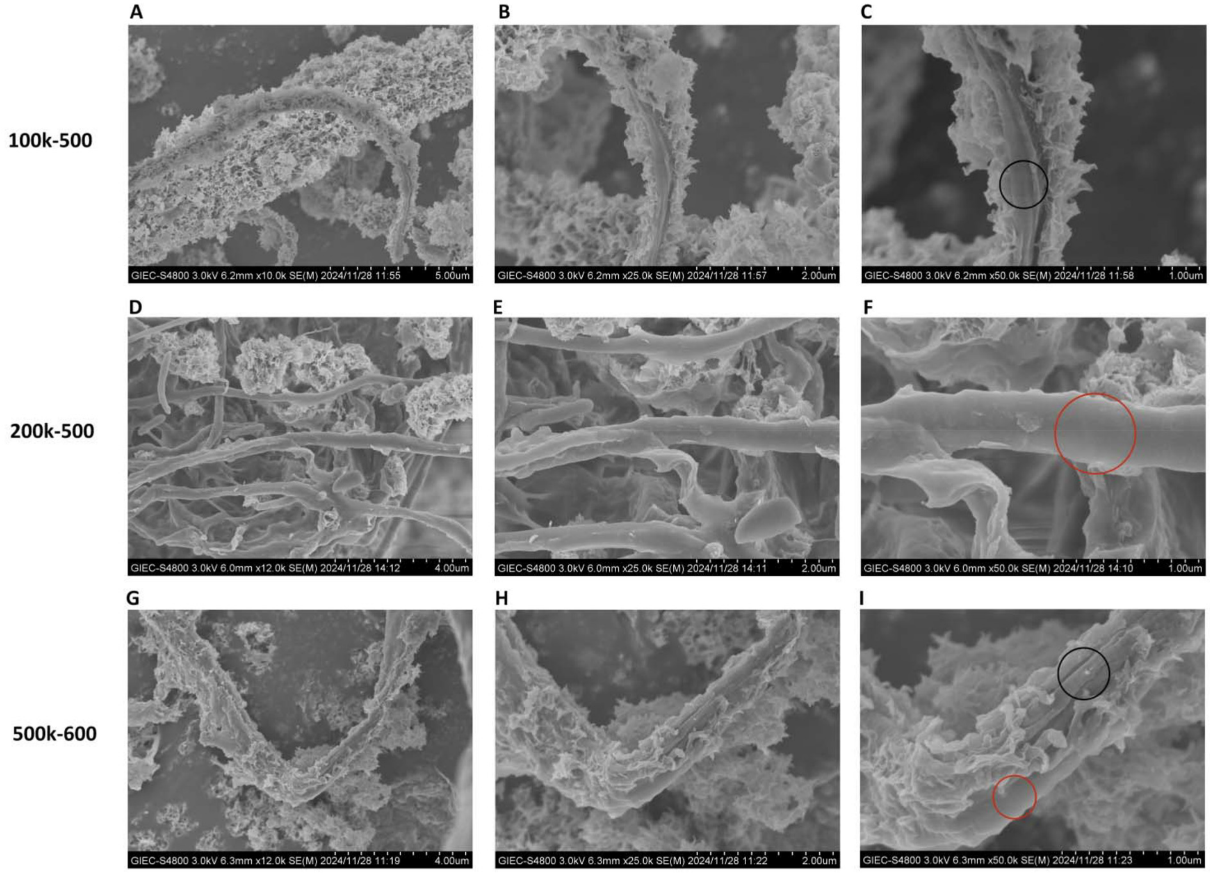

Fig. 5 High power SEM images of 100k–500 (A–C), 200k–500 (D–F) and 500k–600 (G–I) groups. The specific mangification is located at the bottom of each graph. Red circle: collagen with intrafibrillar mineralization; black circle: collagen without intrafibrillar mineralization.

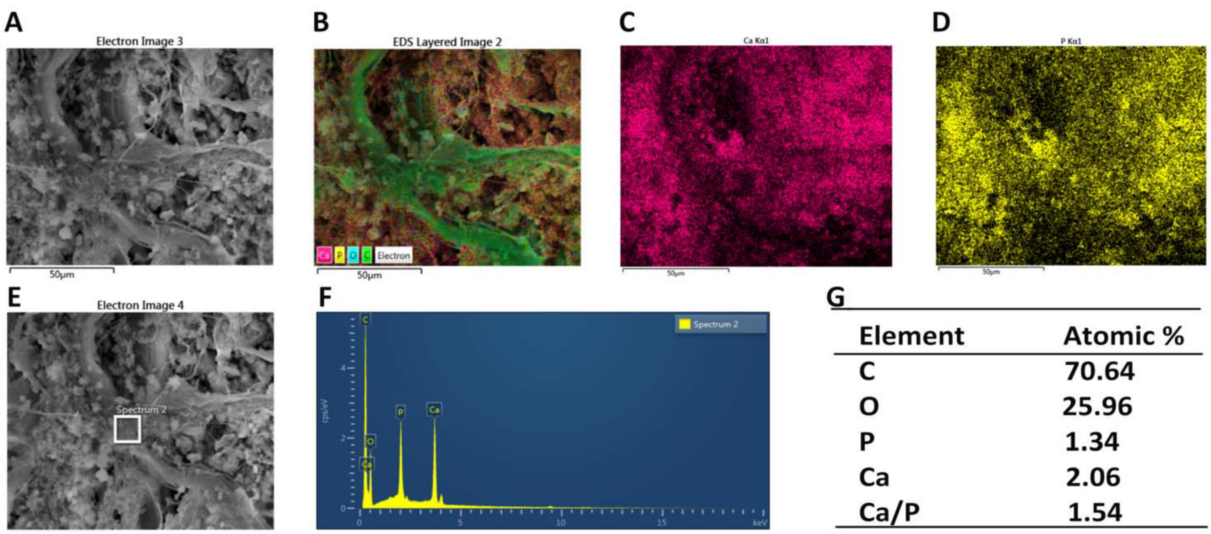

Fig. 6 The mapping result of 200k–500 group (A–D) and point EDS result of collagen in 200k–500 group (E–G). White rectangular box in (E) the range of point scan of EDS.

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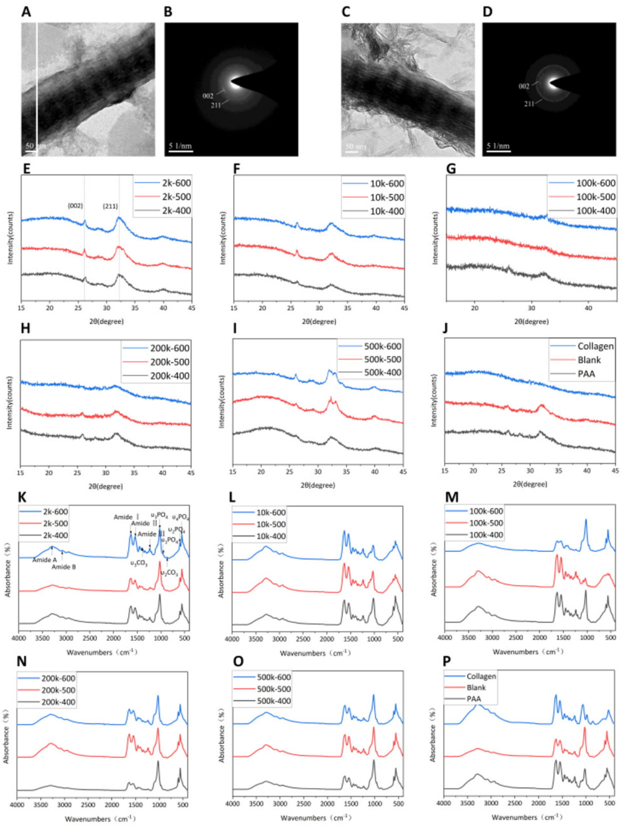

Fig. 7 TEM images and SAED patterns of the 200k–400 group (A and B) and the 200k–500 group (C and D). XRD patterns (E–J) and FTIR spectra (K–P) of the three-dimensional collagen scaffolds in each experimental group after 7 d of mineralization.

The DTG curves for each group (Fig. 8F–J) exhibited consistent descending trajectories, delineated by two distinct weight loss peaks. These peaks signify the progressive elimination of water and organic constituents from the mineralized collagen scaffold as the temperature increased. The initial peak, ranging between 60 and 70 °C, corresponds to the evaporation of water molecules residing within collagen fibres, while the subsequent peak, occurring at 330 to 350 °C, denotes the thermal decomposition of the organic matrix of collagen. Remarkably, both the experimental and positive control groups showed analogous thermal degradation profiles, indicative of a uniform mineral composition within the collagen scaffold.

4. Discussion

This study explored the relationship between the stability of PGA-ACP solution and the occurrence of intrafibrillar mineralization of type I collagen. To assess the stability of PGA-ACP solutions across various molecular weights and concentrations, the changes in the OD value were monitored over a 72 h period and the $t_{1/2}$ were calculated. The results revealed that solutions containing 2 kDa, 10 kDa, 200 kDa, and 500 kDa γ-PGA at concentrations of 400 μg mL^{-1}, 500 μg mL^{-1}, and 600 μg mL^{-1} maintained the stability of supersaturated CaP solutions for a minimum of nearly 10 h. Specifically, the MW of PGA arranged in ascending order of stabilization ability are as follows: 2 kDa < 10 kDa ≤ 500 kDa < 200 kDa. In 200 kDa groups, the higher the concentration, the better the ability of stabilization. Differently, this variation in other group is too slight to confirm. However, the $t_{1/2}$ was smallest and the solutions containing 100 kDa γ-PGA failed to maintain the stability of supersaturated calcium phosphate at concentrations of 400 μg mL^{-1}, 500 μg mL^{-1} and 600 μg mL^{-1}. Simultaneously, the CaP solutions of 2k–600, 10k–600 and 100k–600 groups were directly observed under TEM microscope. To investigate the potential of different PGA-ACP solutions to induce intrafibrillar mineralization, in this study, a two-dimensional collagen model was subjected to mineralization within each PGA-ACP dispersion system for 3 d. TEM images, depicted below, revealed that intrafibrillar mineraliz-

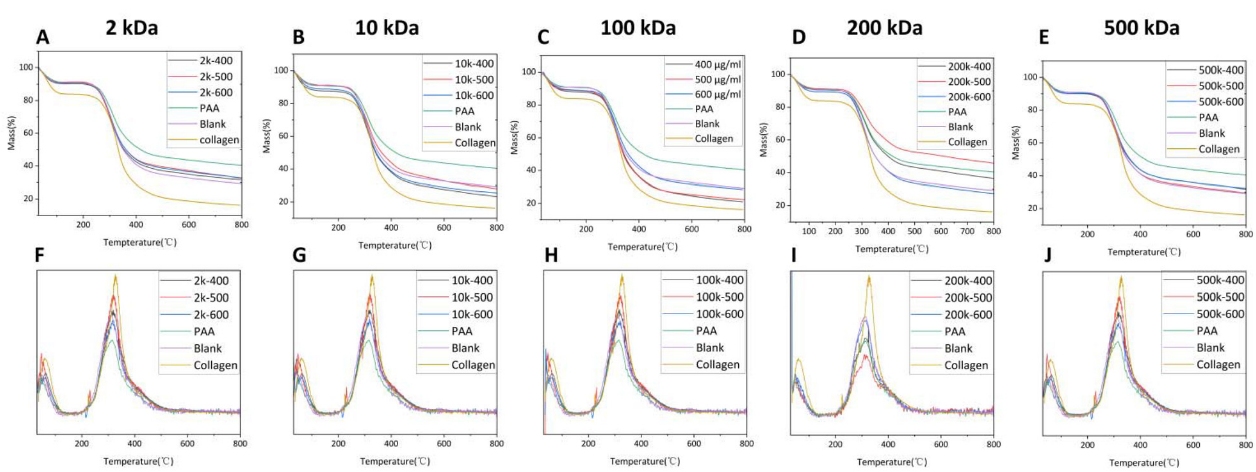

Fig. 8 TG curves (A–E) and DTG analysis curves (F–J) of the three-dimensional collagen scaffold mineralized under the induction of PGA with molecular weights of 2 kDa (A and F), 10 kDa (B and G), 100 kDa (C and H), 200 kDa (D and I) and 500 kDa (E and J) at different concentrations.

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ation occurred within all the groups characterized by stable PGA-ACP solution compositions. Conversely, the experimental groups lacking stability failed to induce intrafibrillar mineralization. And in all the groups with intrafibrillar mineralization, only 200k–400 and 200k–500 groups can find hierarchical intrafibrillar mineralized collagen. The crystallization nature of the mineralization within collagen fibres was confirmed. In summary, the findings of this study demonstrate that the sustained stability of PGA-ACP solution for a minimum of 10 h facilitates intrafibrillar mineralization of the two-dimensional collagen model.

It can't be ignored that the particle size of the 100k–400 group larger than that of blank in 24 h (Fig. 1M), the longer the remineralization time, the less the particle size in Fig. 1M (100 kDa). According to the TEM micrograph in Fig. 2, the 100 kDa keeps a condition between ACP and HA like Fig. 2M–P for at least 24 h, which is a filamentous substance contact with each other might cause bigger particle size than crystal in blank group (ESI Fig. 1†). In the 48 h and 72 h of 100 kDa groups, precipitation visible to the naked eye can be observed, signifying the success of crystallization. In general, the PGA with 100 kDa can stabilize the CaP solutions in a condition between ACP and HA for a long time. During this period, there is a hypothesis that filamentous mineralization precursor aggregations of CaP become bigger and result in similar even higher particle size than blank group. With the time going, this filamentous mineralization precursor aggregations gradually crystallized and show a less particle size than before.

In the study of Shuqin Jiang,[39] the pH curve can divide the crystallization process of ACP into three parts, including induction perio, crystallization stage, snd post crystallization stage. However, in our experimental groups, PGA was used as NCPs analogues, which show lower ability of stabilization than other polyelectrolyte like PAA,[31] pAsp,[29] CMC[32] and so on. Because of this, the PGA-ACP dispersion system might haven't obvious demarcation of the induction stage and crystalization stage which result in irregulaar pH change. The filamentous substance in Fig. 2M–P and ESI Fig. 1D† without characteristic diffraction ring mean it is a stage between ACP and HA. Different from the spherical ACP particle size nearly 50 nm, the filamentous substance are bigger and can not into the collagen fibres. Consequently, the intrafibrillar mineralization is limited. It seems like the 100 kDa groups have idealized pH curve, but the EDS result in Fig. 2D suggests that even the period looks like ACP in 100k–6 h (Fig. 2L) may have begun to crystallize. Therefore, for PGA-ACP dispersion systems, the change in pH value is not of much reference significance.

TEM images only provide insight into the presence of intrafibrillar mineralization and cannot be used to gauge the extent of mineralization induced by PGA. In the results of TEM observation (Fig. 3), it's worth noting that the appearence of hierarchical intrafibrillar mineralization in 200k–400 and 200k–500 groups. Longitudinal comparsion of all the groups with different MW, the 200 kDa groups have strongest stability. Horizontally comparsion of 200 kDa groups with different concentrations, the 500 and 400 μg ml^{-1} aren't the most stable. This remind us might only a specific range of stability can lead to the generation of hierarchical intrafibrillar mineralization.

Considering the three-dimensional scaffold nature of dentin, bone tissue, and other hard tissues predominantly composed of collagen, three-dimensional collagen scaffolds were employed in this study for a 7 d mineralization period to investigate the influence of the molecular weight and concentration of PGA on the degree of mineralization. Compared with the clear intrafibrillar mineralization of the two-dimensional collagen model, the extrafibrillar mineralization of the partially mineralized three-dimensional collagen scaffold is more obvious. And the three-dimensional structure also makes it difficult to determine whether intrafibrillar mineralization has occured. Usually, the thickening collagen fibres indicate the occurrence of intrafibrillar mineralization.[31] Nonetheless, a more pronounced observation within the scope of this experiment pertains to the comparative analysis of the collagenous architecture. In contrast to the loosely arranged collagen fibres found in pristine collagen matrices, the structural configuration of the fibres in the group deemed capable of intrafibrillar mineralization appears to either dissipate or become less discernible. Instead, they coalesce into a contiguous collagen fiber bundle that is distinctly visible under reduced magnification like the PAA group (ESI Fig. 3C and F†). Except 100 kDa groups, both the changed collagen fibre bundle and extrafibrillar mineral substance can be observed in all the other experimental groups (Fig. 4). Interestingly, the surfaces of collagen bundles in 200 kDa group (Fig. 4J–L) rarely had scaly mineralized components as in other groups, but a relatively regular spherical mineralization particles can be seen in 200k–600 and 200k–500 groups. Consequently, the 200k–500 group was chosen as a exemplar for EDS analysis (Fig. 6), in order to ascertain the presence of intrafibrillar mineralization, obviating the influence of minerals adhering to the collagenous surface. Based on the EDS results, the C and O elements intrinsic to the collagen within the collagenous region are distinctly prominent, whereas Ca and P signals are not as evident. However, when the position of the collagen within the mapping image was selected for point scanning, Ca and P signals were still detectable as the strongest after C and O (Fig. 6F). Although the ratio of Ca/P (1.54) is slightly different from that of HA (1.67), the XRD and FTIR analyses of the three-dimensional collagen scaffolds post-mineralization in each experimental group confirm the presence of HAp minerals.

In this study, γ-PGA served as a noncollagenous analogue for stabilizing a supersaturated calcium phosphate solution, and intrafibrillar mineralization was observed to occur when the solution remained stable. Within the realm of biomimetic mineralization, scholars have utilized various polymers *in vitro* as noncollagenous analogues to stabilize supersaturated calcium phosphate solutions and facilitate intrafibrillar mineralization. Qi Y *et al.*[31] employed PAA with molecular weights of 2 kDa, 45 kDa, and 500 kDa at concentrations varying between

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10 μg mL^{-1}, 25 μg mL^{-1}, and 50 μg mL^{-1} to stabilize supersaturated calcium phosphate solutions. The findings indicated that except for the 2k–50 group, in which excessive solution stabilization hindered both intrafibrillar and extrafibrillar mineralization, the groups effectively stabilized the supersaturated calcium phosphate solution for a duration and induced intrafibrillar mineralization of collagen. In contrast, the γ-PGA utilized in our study exhibited a limited ability to stabilize supersaturated calcium phosphate solutions, as evidenced by the increase in the OD observed after a maximum of 10 h and the restricted $t_{1/2}$. Consequently, there was no experimental group in which excessive stability hindered crystallization of calcium and phosphorus ions in collagen fibres. Wang R *et al.*[32] employed carboxymethyl chitosan (CMC) with molecular weights of 20 kDa, 60 kDa, and 150 kDa at concentrations ranging from 50 μg mL^{-1} to 200 μg mL^{-1} to stabilize supersaturated calcium phosphate solutions. The findings revealed that except for the 20k–50 group, which exhibited weaker solution stabilization leading to only extrafibrillar mineralization, the groups effectively stabilized the supersaturated calcium phosphate solution for a duration and facilitated intrafibrillar mineralization. The 100 kDa PGA utilized in our study displayed behaviour akin to that of the 20k–50 group of CMC across diverse concentrations. This similarity was underscored by an early rise in the OD value upon experiment initiation, indicating an inadequate stabilization capability. Consequently, no intrafibrillar mineralization was observed. Furthermore, PAH with a molecular weight of 15 kDa (ref. 22) and PAsp with molecular weights of 1.4 kDa (ref. 29) and 14 kDa,[22] among others, were found to effectively stabilize supersaturated calcium phosphate solutions over a duration and trigger mineralization within collagen fibres. Additionally, there have been no reports of intrafibrillar mineralization occurring in the presence of added polyelectrolytes serving as NCP analogues, in which case the mineral ions are not stabilized. In summary, when utilizing NCP analogues to stabilize supersaturated calcium phosphate solutions for biomimetic mineralization, we believe that a relatively stable supersaturated calcium phosphate solution—neither excessively inhibiting calcium phosphate crystallization within collagen fibres nor exhibiting an inadequate stabilization capability leading to competitive mineral ion deposition outside collagen fibres—is a necessary and sufficient condition for intrafibrillar mineralization to occur.

The use of NCPs and their analogues is not the only way to stabilize supersaturated calcium phosphate solutions to produce ACP. Du T *et al.*[40] employed periodic fluid shear stress (PFSS) to stabilize supersaturated calcium phosphate solutions and observed the formation of mineralization precursors using TEM. In contrast to the ability of PAA to stabilize mineralization fluid to facilitate intrafibrillar mineralization, fluid shear stress was found to enhance the conversion of ACP into apatite crystals, enabling successful observation of intrafibrillar mineralization under TEM. This implies that when alternative methods are employed to stabilize supersaturated calcium phosphate solutions, the results align with the hypothesis that a relatively stable supersaturated calcium phosphate solution is both a necessary and sufficient condition for intrafibrillar mineralization.

Both the polyelectrolyte-ACP dispersion system and the PFSS-ACP dispersion system represent biomimetic mineralization systems grounded in the principles of nonclassical crystallization theory. However, the distinction between nonclassical and classical crystallization theories remains unclear. Thermodynamically stable prenucleation clusters (PNCs) viscosify into hydrated ACP, which serves as a connecting link between classical and nonclassical crystallization theories.[41,42] Following the implementation of semipermeable membrane separation to establish an environment solely utilizing PNCs for guiding intrafibrillar mineralization, TEM and molecular dynamics simulations confirmed the efficacy of PNCs in directing intrafibrillar mineralization. Nonetheless, their efficiency is compromised by the competitive deposition of mineral ions outside collagen fibres.[43] This observation implies that when a sufficiently stable environment is established that enables the penetration of calcium and phosphorus ions into the interior of collagen fibres, irrespective of the entry mechanism, the potential for intrafibrillar mineralization emerges. This phenomenon may be attributed to the inherent template function of collagen fibres themselves.

Sliver and Landis[44] proposed that the polypeptide stereochemistry of collagen plays a role in the orderly arrangement of charged amino acids. This arrangement facilitates attraction and binding of individual calcium and phosphate ions to the negative and positive side chains of the corresponding residues, respectively. Yan Wang *et al.*[28] utilized fibrillar collagen scaffolds generated through the spontaneous self-assembly of collagen molecules at high concentrations, which closely resembled those found in living tissues. They observed the formation of carbonated HAp, akin to the mineral observed in bone, within mimicked extracellular fluid without the presence of NCPs or their analogues. Although the impact of NCPs was disregarded in these investigations, the template function of collagen remains undeniable.

Nowadays, the prevailing theoretical paradigm posits that non-collagenous proteins are instrumental in regulating the mineralization process within collagenous fibers. The utilization of polymer-induced *in vitro* mineralization to mimic non-collagenous proteins plays a pivotal role in the suppression of nucleation and the enhancement of the stabilization of amorphous phases within the solution.[45] Subsequently, the functionality was meticulously examined through the reconstitution of distinct non-collagenous proteins, like dentine phosphorylated protein and dentine matrix protein-1. This recombination proteins enhanced the charge density and structural stability *via* phosphorylation. Specifically, the latter protein facilitated the stabilization of calcium and phosphorus ions at the N terminus, thereby promoting the precipitation of amorphous calcium phosphate. Concurrently, the C terminal contributed to the acceleration of nucleation.[46] Therefore, the concurrent presence of diverse non-collagenous proteins within the organism is apt to modulate the inhibitory threshold of

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nucleation within the ambient milieu *via* phosphorylation or dephosphorylation mechanisms, thereby facilitating the attainment of what was referred to as "relative stable supersaturated calcium phosphate solution".

In summary, this study refrains from delving into the specific driving forces behind calcium and phosphate ion interactions, focusing instead on elucidating the commonalities among various theories of biomineralization. We utilized the PGA-ACP dispersion system and confirmed a direct correlation between the stability of the PGA-ACP solution and the occurrence of intrafibrillar mineralization. Once PGA successfully stabilized the supersaturated calcium phosphate solution for at least 10 h, intrafibrillar mineralization of collagen occurred. Building upon existing research and considering other relevant studies, we expand upon the hypothesis that a relatively stable supersaturated calcium phosphate solution serves as both a necessary and sufficient condition for intrafibrillar mineralization in the polyelectrolyte-ACP stabilization system, the PFSS-ACP stabilization system, and other classic crystallization fields. And within this range, there is the possibility of hierarchical intrafibrillar mineralization occuring at a certain stability. We emphasize the template function of collagen fibres, which significantly contributes to the formation of intrafibrillar mineralization. Simultaneously, we highlight that a relatively stable environment of calcium and phosphate ions is a prerequisite for effective operation of the template function. This hypothesis may offer a novel perspective on biomimetic mineralization and the synthesis of related materials. However, the experimental data in this study are based solely on the PGA-ACP stabilization system, and sufficient evidence to validate this hypothesis is lacking. Therefore, future studies should incorporate additional methods that maintain the stability of calcium and phosphate ions, apart from the use of polyelectrolytes or PFSS, to further investigate this hypothesis.

5. Conclusion

This study revealed a direct correlation between the stability of the supersaturated calcium phosphate solution in a PGA-ACP dispersion system and the occurrence of intrafibrillar mineralization. Specifically, when PGA successfully stabilizes the supersaturated calcium phosphate solution for at least 10 h, intrafibrillar mineralization of collagen fibres occurs, whereas failure to stabilize the solution results in no intrafibrillar mineralization. Under specific stability conditions, such as within the 200k–400 and 200k–500, the occurrence of hierarchical intrafibrillar mineralization is possible. Building upon this finding and considering other scholars' research, we extend the conjecture that a relatively stable supersaturated calcium phosphate solution is a necessary and sufficient condition for intrafibrillar mineralization to the realms of polyelectrolyte-ACP stabilization systems, PFSS-ACP stabilization systems, and classical crystallization theory. We emphasize the role of the template function of collagen fibres in intrafibrillar mineralization, provided that calcium and phosphate ion competitive deposition outside the fibres is avoided, thus offering a direction for future research on biomimetic mineralization. However, since the experimental study only utilized PGA as a NCP analogue to stabilize the supersaturated calcium phosphate solution and lacked alternative methods for stabilizing calcium and phosphate ions, further validation is still required.

Author contributions

Yuwen Zhang and Tong Chen: thematic conception, data curation, formal analysis, investigation, methodology, writing – original draft, and writing – review and editing. Rui Yuan and Yina Cao: conceptualization, writing – review and editing. Qinghui Zhi and Lisha Gu: conceptualization, supervision, validation, writing – review and editing. Huancai Lin: conceptualization, funding acquisition, project administration, resources, supervision, validation, and writing – review and editing.

Data availablity

Data for this article, including original data and figures of manuscript and ESI† are available at Science Data Bank at [**https://www.scidb.cn/en/anonymous/VkpKZllu**].

Conflicts of interest

No conflicts of interest are declared by the authors.

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